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4. As long as an electron revolves in the fixed stationary orbit, it doesn't lose its energy. However, when an electron jumps from higher energy state (E_2) to a lower energy state (E_1), the excess energy is emitted as radiation. The frequency of the emitted radiation is

$$E_2 - E_1 = h\nu$$

and

$$\nu = \frac{(E_2 - E_1)}{h} \text{ -------- (2.2)}$$

Conversely, when suitable energy is supplied to an electron, it will jump from lower energy orbit to a higher energy orbit.

Applying Bohr's postulates to a hydrogen like atom (one electron species such as H, He^+ and Li^{2+} etc..) the radius of the n^{th} orbit and the energy of the electron revolving in the n^{th} orbit were derived. The results are as follows:

$$r_n = \frac{(0.529)n^2}{Z} A \text{ ---------- (2.3)}$$

$$E_n = \frac{(-13.6)\,Z^2}{n^2} \text{eV atom}^{-1} \text{ --------(2.4)}$$

(or)

$$E_n = \frac{(-1312.8)\,Z^2}{n^2} \text{kJ mol}^{-1} \text{ --------(2.5)}$$

The detailed derivation of r_n and E_n will be discussed in 12^{th} standard atomic physics unit.

2.1.2 Limitation of Bohr's atom model:

The Bohr's atom model is applicable only to species having one electron such as hydrogen, Li^{2+} etc... and not applicable to multi electron atoms. It was unable to explain the splitting of spectral lines in the presence of magnetic field (Zeeman effect) or an electric field (Stark effect). Bohr's theory was unable to explain why the electron is restricted to revolve around the nucleus in a fixed orbit in which the angular momentum of the electron is equal to $nh/2\pi$ and a logical answer for this, was provided by Louis de Broglie.

2.2 Wave particle duality of matter

Albert Einstein proposed that light has dual nature. i.e. light photons behave both like a particle and as a wave. Louis de Broglie extended this concept and proposed that all forms of matter showed dual character. To quantify this relation, he derived an equation for the wavelength of a matter wave. He combined the following two equations of energy of which one represents wave character ($h\upsilon$) and the other represents the particle nature (mc^2).

(i) Planck's quantum hypothesis:

$$E = h\nu \text{ -------- (2.6)}$$

(ii) Einsteins mass-energy relationship

$$E = mc^2 \text{ ---------(2.7)}$$

From (2.6) and (2.7)

$$h\nu = mc^2$$

$$hc/\lambda = mc^2$$